

EG/2020/4291

Wijerathna W.M.M.T.

## **Project Title**

Weak Hashed Password Cracker

## **Project Description**

This project aims to develop a high-performance password-cracking application that recovers numeric-only passwords from their hashed representations (SHA-256). The system will use brute-force techniques to iterate over all possible numeric combinations within a given length and match against a given hash. Due to the massive computational requirement of brute-force cracking, the program will leverage parallel and distributed computing techniques.

## **OpenMP**

- Easier to implement with simple `#pragma` directives.
- No need to manually manage thread creation, joining, or synchronization.
- Ideal for loop-based parallelism (like iterating over password candidates).
- Cleaner, shorter, and more maintainable code.

## **MPI**

- No need for specialized GPU hardware.
- Simpler to learn and implement for a CPU-based brute-force task.
- CUDA requires NVIDIA GPUs